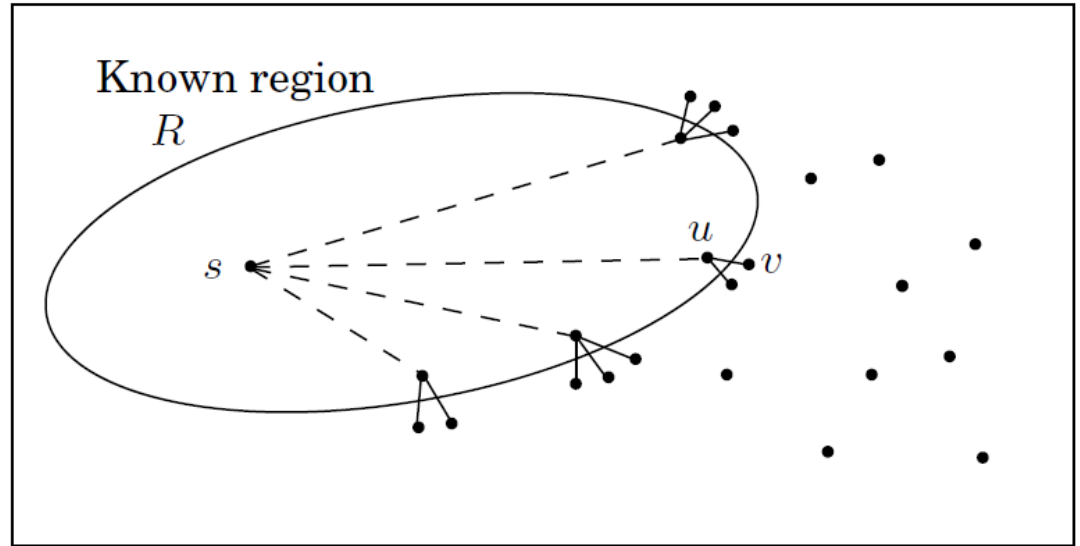
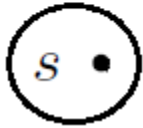


Graphs

Algo run:

Known
region

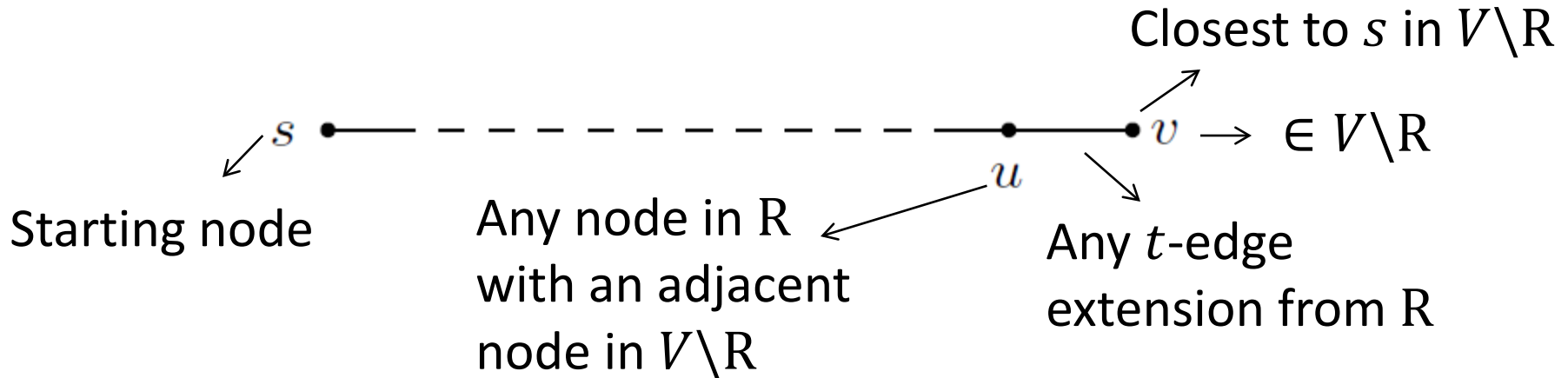
R



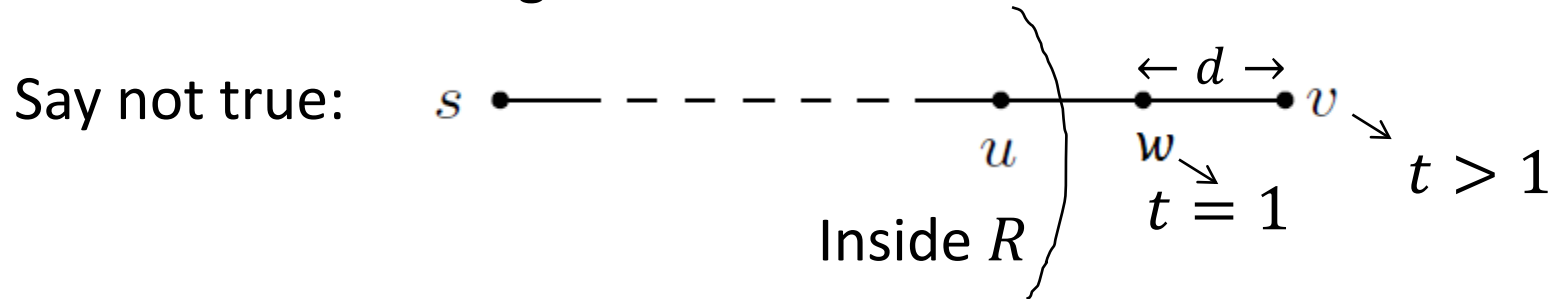
“known region” is some subset of vertices R that includes s



next addition to it should be *the node outside R that is closest to s*



Property: $t = 1$. That is, if v is closest to s in $V \setminus R$ then v is at 1 - edge extension from R .



But as $d > 0$! How can v be the closest to s ?

Only a w at $t = 1$ can, so $v = w$. $\longrightarrow$ Which? shortest (among the extensions!)

Pseudo code:

Initialize $\text{dist}(s)$ to 0, other $\text{dist}(\cdot)$ values to ∞
 $R = \{ \}$ (the 'known region')

while $R \neq V$:

 Pick the node $v \notin R$ with smallest $\text{dist}(\cdot)$

 Add v to R

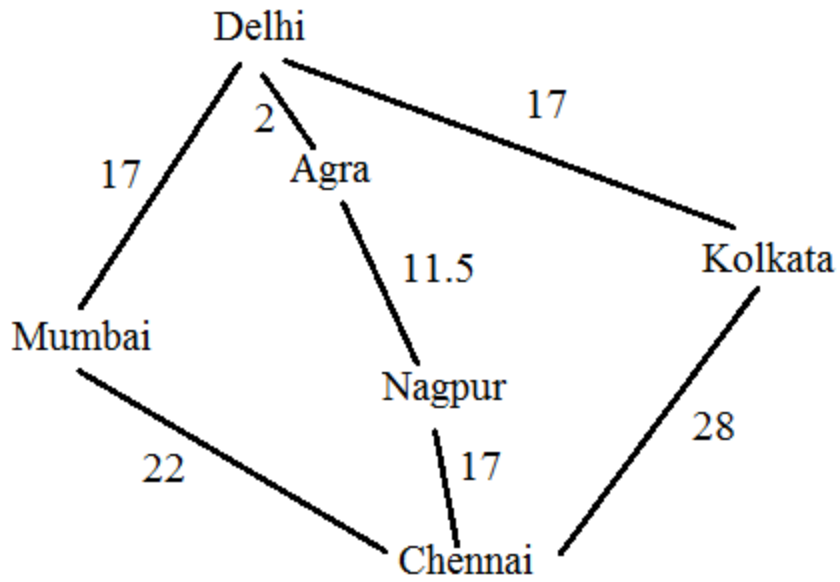
 for all edges $(v, z) \in E$:

 if $\text{dist}(z) > \text{dist}(v) + l(v, z)$:

$\text{dist}(z) = \text{dist}(v) + l(v, z)$

 } Due to the property





1. R: D

M:17,D	N: ∞ ,[]
A: 2,D	C: ∞ ,[]
K:17,D	D: 0, D

2. R: D,A

M:17,D	N:13.5,A
A: 2,D	C: ∞ ,[]
K:17,D	D: 0, D

3. R:
D,A,N

M:17,D	N:13.5,A
A: 2,D	C:30.5,N
K:17,D	D: 0, D

4. R:
D,A,N,K

M:17,D	N:13.5,A
A: 2,D	C:30.5,N
K:17,D	D: 0, D

5. R:
D,A,N,K,
M

M:17,D	N:13.5,A
A: 2,D	C:30.5,N
K:17,D	D: 0, D

6. R:
D,A,N,K,
M,C

M:17,D	N:13.5,A
A: 2,D	C:30.5,N
K:17,D	D: 0, D